

徐圣毅 12310833 HW1

$$2) (1) \vec{A} - \vec{B} = (-1, -1, 5) \quad |\vec{A} - \vec{B}| = \sqrt{(-1)^2 + (-1)^2 + 5^2} = \sqrt{1+1+25} = \sqrt{27} = 3\sqrt{3}$$
$$\vec{u} = \frac{\vec{A} - \vec{B}}{|\vec{A} - \vec{B}|} = \left(-\frac{\sqrt{3}}{9}, -\frac{\sqrt{3}}{9}, \frac{5\sqrt{3}}{9}\right)$$

$$(2) \vec{C} = \vec{A} + \vec{B} + \frac{1}{2}(\vec{A} - \vec{B}) = \cancel{2} (2, 3, -2) + \frac{1}{2}(-1, -1, 5)$$

$$= (2, 3, -2) + \left(-\frac{1}{2}, -\frac{1}{2}, \frac{5}{2}\right) = \left(\frac{3}{2}, \frac{5}{2}, \frac{1}{2}\right)$$

$$\vec{u} = \frac{\vec{C}}{|\vec{C}|} = \left(\frac{3}{\sqrt{35}}, \frac{5}{\sqrt{35}}, \frac{1}{\sqrt{35}}\right)$$

$$3) \vec{B} = \left(\frac{2\lambda}{3}, -\frac{2\lambda}{3}, \frac{\lambda}{3}\right) \quad \vec{A} - \vec{B} = \left(6 - \frac{2}{3}\lambda, 6 - 2 + \frac{2}{3}\lambda, -4 - \frac{1}{3}\lambda\right)$$

$$|\vec{A} - \vec{B}| = \sqrt{\left(6 - \frac{2}{3}\lambda\right)^2 + \left(\frac{2}{3}\lambda - 2\right)^2 + \left(-4 - \frac{1}{3}\lambda\right)^2}$$

$$= \sqrt{\frac{4}{9}\lambda^2 - 8\lambda + \frac{36}{36} + \frac{4}{9}\lambda^2 - \frac{8}{3}\lambda + 4 + 16 + \frac{8}{3}\lambda + \frac{1}{9}\lambda^2}$$

$$= \sqrt{\lambda^2 - 8\lambda + 36} = 10$$

$$\text{so : } \lambda^2 - 8\lambda + 36 = 100 \Rightarrow \lambda^2 - 8\lambda - 44 = 0$$

$$\lambda = \frac{8 \pm \sqrt{64 + 176}}{2} = \frac{8 \pm \sqrt{240}}{2} = \frac{8 \pm 2\sqrt{60}}{2} = \frac{8 \pm 4\sqrt{15}}{2} = 4 \pm 2\sqrt{15}$$

Since direction is same for unit vector and $\vec{OB}$:

$$\lambda > 0, \quad \lambda = 4 + 2\sqrt{15}$$

$$B \left(\frac{8 + 4\sqrt{15}}{3}, -\left(\frac{8 + 4\sqrt{15}}{3}\right), \frac{4 + 2\sqrt{15}}{3} \right)$$

$$4) (1) \vec{G}_p = 24xy \vec{a}_x + 12(x^2+2) \vec{a}_y + 18z^2 \vec{a}_z$$

$$G_{xp} = \cancel{24} 24 \times 1 \times 2 = 48 \quad G_{yp} = 12(1^2+2) = 36 \quad G_{zp} = 18(1)^2 = 18$$

$$\vec{G}_p = 48 \vec{a}_x + 36 \vec{a}_y + 18 \vec{a}_z$$

$$(2) \vec{G}_Q = G_{xQ} = 24(-2 \times 1) = -48 \quad G_{yQ} = 12((-2)^2+2) = 12(4+2) = 72$$

$$G_{zQ} = 18(3)^2 = 18 \times 9 = 162$$

$$\vec{G}_Q = -48 \vec{a}_x + 72 \vec{a}_y + 162 \vec{a}_z$$

$$|\vec{G}_Q| = \sqrt{(-48)^2 + (72)^2 + (162)^2} \approx 183.7 \approx 184$$

$$\vec{u} = \frac{-48}{184} \vec{a}_x + \frac{72}{184} \vec{a}_y + \frac{162}{184} \vec{a}_z$$

$$\approx -0.26 \vec{a}_x + 0.39 \vec{a}_y + 0.88 \vec{a}_z = (-0.26, 0.39, 0.88)$$

$$(3) \vec{OP} - \vec{OQ} = (3, 1, -4) \quad |\vec{OP} - \vec{OQ}| = \sqrt{3^2 + 1^2 + (-4)^2} = \sqrt{9+1+16} = \sqrt{26}$$

$$\vec{u}' = \left(\frac{3}{\sqrt{26}}, \frac{1}{\sqrt{26}}, -\frac{4}{\sqrt{26}} \right)$$

$$(4) |\vec{G}| = \sqrt{(4xy)^2 + [12(x^2+2)]^2 + (18z^2)^2} = \sqrt{576x^2y^2 + 144(x^4+4x^2+4) + 324z^4}$$

$$= \sqrt{576x^2y^2 + 144x^4 + 576x^2 + 576 + 324z^4} = 60$$

$$\text{so : } 576x^2y^2 + 144x^4 + 576x^2 + 576 + 324z^4 = 3600$$

$$\text{equation : } 576x^2y^2 + 144x^4 + 324z^4 + 576x^2 - 3024 = 0$$

$$48x^2y^2 + 12x^4 + 27z^4 + 48x^2 - 252 = 0$$

$$b) \quad (1) \quad \vec{r}_1 \times \vec{r}_2 = \begin{vmatrix} \vec{a}_x & \vec{a}_y & \vec{a}_z \\ 7 & 3 & -2 \\ -2 & 7 & -3 \end{vmatrix} = 5 \vec{a}_x + (-1)(-25) \vec{a}_y + 55 \vec{a}_z$$

$$= (5, 25, 55)$$

$$\vec{a} = (5, 25, 55) \quad |\vec{a}| = \sqrt{5^2 + 25^2 + 55^2} = \underline{\underline{\approx 60.6 \approx 61}}$$

$$\vec{u} = \left(\frac{5}{61}, \frac{25}{61}, \frac{55}{61} \right)$$

$$(2) \quad \vec{r}_1 - \vec{r}_2 = (9, -4, 1) \quad \vec{r}_2 - \vec{r}_3 = (-2, 5, -6)$$

$$(\vec{r}_1 - \vec{r}_2) \times (\vec{r}_2 - \vec{r}_3) = \begin{vmatrix} \vec{a}_x & \vec{a}_y & \vec{a}_z \\ 9 & -4 & 1 \\ -2 & 5 & -6 \end{vmatrix} = 19 \vec{a}_x + (-1)(-52) \vec{a}_y + 37 \vec{a}_z$$

$$\vec{b} = (19, 52, 37) \quad |\vec{b}| = \sqrt{19^2 + 52^2 + 37^2} \approx 66.58 \approx 67$$

$$\vec{u}' = \left(\frac{19}{67}, \frac{52}{67}, \frac{37}{67} \right)$$

$$(3) \quad \vec{r}_1 \times \vec{r}_2 = (5, 25, 55) \quad S = \frac{1}{2} |\vec{r}_1 \times \vec{r}_2|$$

$$S = \frac{1}{2} \times \sqrt{5^2 + 25^2 + 55^2} \approx \frac{61}{2} = 30.5$$

$$(4) \quad (\vec{r}_1 - \vec{r}_2) \times (\vec{r}_2 - \vec{r}_3) = (19, 52, 37)$$

$$S = \frac{1}{2} |(\vec{r}_1 - \vec{r}_2) \times (\vec{r}_2 - \vec{r}_3)| \approx \frac{1}{2} \times 67 = 33.5$$